

Set Theory SoSe 25-26, Wise 25-26

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3 On Hilbert's First Problem

Definition 3.1 (Continuum Hypothesis). The Continuum Hypothesis(CH) asserts that there does not exist an uncountable subset A of $\mathbb{R}$ such that $\overline{\overline{N}} < \overline{\overline{A}} < \overline{\overline{\mathbb{R}}}$, *i.e.*, $\aleph_1 = 2^{\aleph_0}$.

Definition 3.2 (Generalized Continuum Hypothesis). The Generalized Continuum Hypothesis(GCH) asserts that for all $\alpha < \text{Ord}$, $2^{\aleph_\alpha} = \aleph_{\alpha+1}$.

Kurt Gödel in 1938 showed that $\text{Con}(\text{ZFC}) \rightarrow \text{Con}(\text{ZFC} + \text{GCH})$ using his constructible universe(L), but in this section we will give a ZFC model where CH holds by forcing.

3.1 A ZFC model in which CH fails

Definition 3.3. Let $I \neq \emptyset$, define $\mathbb{C}(I) = \{p : p \text{ is a function } \text{dom}(p) \rightarrow \omega, \text{dom}(p) \subseteq I \times \omega, \overline{\overline{\text{dom}(p)}} < \aleph_0\}$, ordered by reverse inclusion, *i.e.*, product of $\overline{\overline{I}}$ -many Cohen forcing with finite support.

Observations: $\mathbb{C}(I)$ has the c.c.c. and $1 \Vdash \overline{\overline{I}} \leq 2^{\aleph_0}$.

Theorem 3.4 (Paul Cohen, 1963). *If ZFC is consistent then so is $\text{ZFC} + \neg\text{CH}$. In fact $\text{Con}(\text{ZFC}) \rightarrow \text{Con}(\text{ZFC} + 2^{\aleph_0} = \aleph_2)$*

Proof. From now on always assume M is a countable transitive model of ZFC. We may assume $M \models \text{CH}$. We force with $\mathbb{C}(\omega_2^M)$. Let $G \subseteq \mathbb{C}(\omega_2^M)$ be a generic filter over M . As $\mathbb{C}(\omega_2^M)$ has the c.c.c. so $\omega_2^M = \omega_2^{M[G]}$.

Since $1 \Vdash \overline{\overline{I}} < 2^{\aleph_0}$, there exists a injection $F : \omega_2 \rightarrow \omega^\omega \cap M[G]$ in $M[G]$ so $M[G] \models 2^{\aleph_0} \geq \aleph_2$ thus $M[G] \models \neg\text{CH}$. Now we show $M[G] \models 2^{\aleph_0} \leq \aleph_2$. Let $x \in \omega^\omega \cap M[G]$ and say $x = \tau^G$, define $E_n = \{p \in \mathbb{C}(\omega_2^M) : \exists m < \omega, p \Vdash \tau(\check{n}) = \check{m}\}$ for $n < \omega$. Since E_n is dense, let $A_n \subseteq E_n$ be an maximal antichain and define $\sigma = \{([\check{n}, \check{m}], p) : p \in A_n, p \Vdash \tau(\check{n}) = \check{m}\}$ where $[\rho, \sigma] = \{(\{(\rho, 1), (\sigma, 1)\}, 1), (\{(\rho, 1)\}, 1)\}$ for $y = \tau^G, z = \sigma^G$, in fact $[\rho, \sigma]^G = (y, z)$. We claim $x = \tau^G = \sigma^G$. If $(n, m) \in \sigma^G$ then $\exists p \in G$ such that $([\check{n}, \check{m}], p) \in \sigma, p \Vdash \tau(\check{n}) = \check{m}$, so $(n, m) \in \tau^G$. For the other side $\tau^G \subseteq \sigma^G$, the proof is similar. So every real $x \in \omega^\omega \cap M[G]$ has a "nice" name. Notice that for any $n < \omega$, there are countably many m and p such that $([\check{n}, \check{m}], p) \in \sigma, \sigma \subseteq \omega \times \omega \times \mathbb{C}(\omega_2^M)$. Thus there are only $(\omega_2^M)^{\aleph_0}$ -many nice names in M , by Hausdorff's formula we obtain: $(\omega_2^{\aleph_0})^M = (2^{\aleph_0} \cdot \omega_2)^M \stackrel{\text{CH}}{=} (\omega_1 \cdot \omega_2)^M = (\omega_2)^M$. Since there are $\omega_2^M = \omega_2^{M[G]}$ many names, $M[G] \models 2^{\aleph_0} = \aleph_2$. ■

However, we are not done yet. We want to get rid of the transitivity hypothesis.

Theorem 3.5 (Levy-Montague Reflection Principle). *Let $\varphi(x, \vec{y}) \in \mathcal{L}_\in$ and $\vec{b}$ be sets, then there is a club class C of ordinals such that for any $\alpha \in C, \vec{b} \in V_\alpha$ and for all $a \in V_\alpha$ we have $V_\alpha \models \varphi(a, \vec{b}) \iff V \models \varphi(a, \vec{b})$.*

Sketch. By induction on the complexity of φ . Let $(\psi_i : i \leq n)$ be an enumeration of subformulae of φ : $\psi_n = \varphi, \psi_i = \psi_k \wedge \psi_j$ if $k, j \leq i$. By induction, we construct c_i : for atomic formula ψ_i ; for negative $\psi_i := \neg\psi_j$ we put $c_i = c_j$; for $\psi_i = \psi_j \wedge \psi_k, c_i = c_j \cap c_k$; for $\psi_i := \exists x \psi_j(x, y, \vec{b})$, let $c_i = \{\alpha \in c_j : \forall y \in V_\alpha, V_\alpha \models \exists x, \psi_j(x, y, \vec{b})\}$. ■

We already showed if M is a countable transitive model of ZFC and G is a generic filter of $\mathbb{C}(\omega_2^M)$ over M then $M[G] \models \text{ZFC} + \neg\text{CH}$. Notice that our proof was finite, so in fact if $\mathcal{E} \subseteq \text{ZFC} + \{\neg\text{CH}\}$ is finite, then there is a finite set $\Gamma(\mathcal{E}) \subseteq \text{ZFC}$ such that if $M \models \Gamma(\mathcal{E})$ where M is a countable transitive model and G is generic then $M[G] \models \mathcal{E}$. Here $\Gamma(\mathcal{E})$ denotes a finite set of axioms of ZFC sufficient to carry out the forcing argument showing that $M[G] \models \mathcal{E}$.

Now we show $\text{Con}(\text{ZFC}) \rightarrow \text{Con}(\text{ZFC} + \neg\text{CH})$:

Suppose $\text{Con}(\text{ZFC})$ which means there exists a model $(N, \in^N)$ of ZFC. By compactness theorem, it suffices to verify $\text{Con}(\mathcal{E})$ for finite $\mathcal{E} \subseteq \text{ZFC} + \neg\text{CH}$. Let $\mathcal{E}$ be as such: By the above paragraph, let $\Gamma = \Gamma(\mathcal{E}) \subseteq \text{ZFC}$. Since " $N \models \text{ZFC}$ " we have $N \models \exists \alpha, V_\alpha \models \Gamma$. Now in N use the Löwenheim-Skolem theorem (and a possible collapse) to get M in N such that $N \models "M \text{ is a countable transitive model of } \Gamma"$.

Also from $N \models \text{ZFC}$ we have $N \models "M \models \Gamma(\mathcal{E}) \text{ and } M[G] \models \mathcal{E}"$. Let $G \in N$ be a generic $\mathbb{C}(\omega_2^M)$ -filter over M , then $N \models "M[G] \models \mathcal{E}"$ thus $N \models \text{Con}(\mathcal{E})$. Since $\text{Con}(\mathcal{E})$ is a Π_1^0 statement about the natural number, but $\omega^V \subseteq \omega^N$ then we have $(\omega^V, 0, 1, +, \cdot) \models \text{Con}(\mathcal{E})$. But then $\mathcal{E}$ is really consistent, which gives us $\text{Con}(\text{ZFC} + \neg\text{CH})$. This completes the proof.

Remark 3.6. By replacing the forcing $\mathbb{C}(\omega_2^M)$ and $\neg\text{CH}$, this procedure can be done for any $\mathbb{P}$ and φ . So from now on, for $\text{Con}(-)$ we may and shall work with transitive models. In fact, we will assume we are forcing over the whole universe V and that there are generics somewhere outside.

3.2 A ZFC model in which CH holds

Idea: Add a Cohen subset of ω_1 . Let $\mathbb{C}_{\omega_1} := \mathbb{C}_{\omega_1}(1) := \text{Add}(\omega_1, 1) = {}^{<\omega_1}\omega_1$ and order it by inverse of end-extension (we use the Cohen forcing at ω_1 .)

Notice that $\overline{{}^{<\omega_1}\omega_1} = \Sigma_{\alpha < \omega_1} \overline{\alpha} \omega_1 = \Sigma_{\alpha < \omega_1} \omega_1^{\aleph_0} = \omega_1 \cdot \omega_1^{\aleph_0} = \omega_1 \cdot \omega_1 \cdot 2^{\aleph_0} = 2^{\aleph_0}$. Thus $\mathbb{C}_{\omega_1}$ has the $2^{\aleph_0} + \text{-c.c.}$

Now we fix $G \subseteq \mathbb{C}_{\omega_1}$ a generic filter over V .

Definition 3.7. Let $\mathbb{P}$ be a poset and κ a regular cardinal, we say $\mathbb{P}$ is $< \kappa$ -closed iff $\forall \lambda < \kappa$ and $\forall (p_\alpha : \alpha < \lambda)$ such that $p_\alpha \in \mathbb{P}$ for all α and for all $\alpha < \beta < \lambda, p_\beta \leq p_\alpha$, there exists $q \in \mathbb{P}$ such that $q \leq p_\alpha$ for all $\alpha < \lambda$.

Lemma 3.8. Let $\mathbb{P}$ be a $< \kappa$ -closed poset and $g \subseteq \mathbb{P}$ be generic. Then ${}^{<\kappa}V \cap V = {}^{<\kappa}V \cap V[g]$ thus it also preserves cardinals $\leq \kappa$.

Proof. Let $\lambda < \kappa, f : \lambda \rightarrow V$ and say that $f = \tau^g \in V[g]$. We want to show $f \in V$. Let $X \in V$ be such that $\text{ran}(f) \subseteq X$ and let $p \in g$ such that $p \Vdash \tau : \check{X} \rightarrow \check{X}$ is a function". Say $q \in \mathbb{P}$ decides τ if and only if for all $\alpha < \lambda, \exists b \in X$ such that $q \Vdash \tau(\check{\alpha}) = \check{b}$.

We want to show that $D = \{q \in \mathbb{P} : q \text{ decides } \tau\}$ is dense below p : take any $r \leq p$ and we construct a sequence $(q_\alpha : \alpha \leq \lambda)$ as follows:

- set $q_0 = r$,
- suppose q_α has already been constructed, define $q_{\alpha+1} \leq q_\alpha$ by $\exists b \in X$ such that $q_{\alpha+1} \Vdash \tau(\check{\alpha}) = \check{b}$, by fullness, such a condition $q_{\alpha+1}$ existed,

- at limit step, by $< \kappa$ -closedness, we set $q_\alpha \leq q_\beta$ for limit α and $\forall \beta < \alpha$.

Now $q_\lambda \leq r$ decides τ . Thus $q \in D \cap G \neq \emptyset$, let $f_q: \lambda \rightarrow X$ be the function that $q \Vdash f_q(\check{\alpha}) = \tau(\check{\alpha})$ and thus $f = f_q \in V$. ■

Note that $\mathbb{C}_{\omega_1}$ is $< \aleph_1$ -closed as ω_1 is regular thus $\omega_1^V = \omega_1^{V[G]}$ and ${}^\omega\omega \cap V = {}^\omega\omega \cap V[g]$. In order to show $\text{Con}(\text{ZFC} + \text{CH})$, we need to show $V[G] \models 2^{\aleph_0} = \aleph_1$, i.e., $V[G] \models \exists a$ surjection $\omega_1 \rightarrow \omega^\omega$.

Proof. Let $F = \bigcup G: \omega_1 \rightarrow \omega_1$, now we construct a surjective function $\hat{F}: \omega_1 \rightarrow \omega^\omega$ in $V[G]$.

Define

$$\hat{F}(\alpha)(n) = \begin{cases} F(\omega \cdot \alpha + n) & \text{if } F(\omega \cdot \alpha + n) < \omega \\ 0 & \text{otherwise} \end{cases}$$

We claim that $\hat{F}$ is onto: let $x \in \omega^\omega$, define $D_x = \{p \in \mathbb{C}_{\omega_1} : \exists \alpha < \omega_1, \forall n < \omega, p(\omega \cdot \alpha + n) = x(n)\} \in V$, it is easy to see D_x is dense. Let $p \in D_x \cap G$, then there exists $\alpha < \omega_1$ such that for any $n < \omega$, $p(\omega \cdot \alpha + n) = x(n) = F(\omega \cdot \alpha + n) = F(\omega \cdot \alpha + n) = \hat{F}(\alpha)(n)$ thus $\hat{F}(\alpha) = x$, $\hat{F}$ is onto.

Therefore $V[G] \models \text{CH}, \text{Con}(\text{ZFC}) \rightarrow \text{Con}(\text{ZFC} + \text{CH})$. ■

We have settled the continuum problem.